



FOOD PRESERVATION, NUTRITION & MEDICINE — FULL MIND MAP

Chemistry · Chapter 8 · Topic-wise · Fully Explained · Hinglish · SSC/PCS/UPSC Ready

Topics in this chapter:

1. Food Preservation, Refrigeration & Additives
2. Sugars, Sweeteners & Nutrition Chemistry
3. Medicinal Chemistry & Aspirin
4. Antibiotics, Drug Classes & Sedatives
5. Toxicology, Food Processing & Organic Food

1. Food Preservation, Refrigeration & Additives

- Fruits/vegetables par wax emulsion coating storage life badhane ke liye use hoti hai; ye moisture loss aur respiration ko limit kar sakti hai aur preservatives/growth regulators ka carrier bhi ban sakti hai. Galat coating se off-flavour develop ho sakta hai. [Q1]
- Sodium benzoate ($C_7H_5NaO_2$, E211) acidic foods/fruit juices me common chemical preservative hai. Benzoic acid aur uske salts microbial growth ko inhibit karte hain; isi liye food preservation questions me repeatedly pooche gaye hain. [Q2–Q6]
- Potato chips ko Nitrogen atmosphere me pack kiya jata hai taaki oxidation/rancidity kam ho, moisture se protection mile aur bag cushioning de. Cold storage me respiration aur ethylene production ki rate kam hoti hai, isliye fruits ki ripening slow aur shelf life longer hoti hai. [Q7, Q9]
- MSG (Monosodium glutamate/Ajinomoto) glutamic acid ka sodium salt hai aur food me mainly umami/flavour enhancer ke roop me use hota hai. Source question ise flavour enhancer ke saath preserver bhi mark karta hai, isliye exam context me multiple-answer note important hai. [Q8]
- Refrigeration food ko sterilize nahi karti; ye microorganisms ki growth aur biochemical reactions ko slow karti hai. Source-era domestic-refrigerator question Freon ko common refrigerant batata hai. Triclosan antibacterial chemical toiletries jaise soaps/toothpastes/cosmetics me mil sakta hai. [Q10–Q12]
- Animal/bone charcoal historically sugar refining me decolourizing/de-ashing agent ke roop me use hua hai. Activated carbon-type adsorbents coloured impurities ko surface par adsorb karke clarification me help karte hain. [Q13]

EXAM TRAP — Q8 & Q11 — Source wording vs modern chemistry

- ▶ Q8 source MSG ko flavour enhancer + preserver dono mark karta hai. Standard food-science role mainly flavour enhancer (umami) hai; “preservative” ko general primary function mat samjho.
- ▶ Q11 ka Freon answer old-exam context hai. Modern domestic refrigerators me ozone-depleting CFC Freons largely phase-out ho chuke hain; newer refrigerants (e.g. R-600a) common hain.

2. Sugars, Sweeteners & Nutrition Chemistry

- Honey me fructose aur glucose major sugars hain; source composition me fructose sabse bada fraction diya hai. Natural common sugars me Fructose sabse sweet hota hai. Relative naming: grape sugar = glucose, fruit sugar = fructose, milk sugar = lactose, table sugar = sucrose, Stevia = natural sweetener. [Q14–Q17]
- Sugar beet → sugar, honey → glucose + fructose, cotton → cellulose aur milk → lactose — ye chaaron pairs correctly matched hain. Invert sugar sucrose ke hydrolysis se glucose + fructose deta hai; free fructose ke karan sucrose se sweeter hota hai. [Q16, Q18]
- Aspartame approx. 200× sucrose se sweeter hai; isliye bahut small quantity me desired sweetness milti hai aur calorie contribution low ho jata hai, even though gram-for-gram energy deta hai. Given artificial sweeteners me source ke according Alitame (~2000× sucrose) highest sweetness value deta hai. [Q19–Q20]

3. Medicinal Chemistry & Aspirin

- Medicinal chemistry me drugs ko molecular targets ke basis par classify karna highly useful hai, kyunki drugs proteins/nucleic acids etc. target biomolecules ke saath interact karke specific mechanism of action dikhati hain. [Q21]
- Aspirin = acetylsalicylic acid; salicylate chemistry ka historical origin willow bark se linked hai. Aspirin analgesic, antipyretic aur anti-inflammatory drug hai; given options me antipyretic classification accepted hai. [Q22–Q23]

EXAM TRAP — Q22 — “Aspirin is obtained from a tree”

- ▶ Historical natural precursor salicylic acid willow bark se mila tha; modern aspirin directly tree se extract nahi hota, industrially salicylic acid ki acetylation se synthesize hota hai.
- ▶ Exam option me “a tree” historical-origin sense me accepted hai.

4. Antibiotics, Drug Classes & Sedatives

- Natural antibiotics ke important producers soil microorganisms hain—especially Streptomyces/actinomycetes (bacteria) aur kuch fungi. Source Q24 “mostly bacteria” ko key karta hai because known natural antibiotics ka large share bacterial/actinomycete origin ka hai. [Q24]

- Penicillin antibiotic hai. Source Q25 me Penicillin aur Sulfadiazine dono antibacterial options hone ki wajah se “more than one” diya gaya hai; Q26 me Penicillin clearly antibiotic option hai. Aspirin/Paracetamol analgesic-antipyretic aur Chloroquine antimalarial category me aate hain. [Q25–Q26]
- Drug-class matching: Paracetamol = antipyretic; Calciferol (Vitamin D) = antirachitic; Aspirin antiseptic nahi hai. Chloromycetin = antityphoid use, Crystal violet = antiseptic, Quinine = antimalarial; Aspirin anaesthetic nahi, analgesic/antipyretic hai. [Q27–Q28]
- Potassium bromide (KBr) historically sedative aur anticonvulsant ke roop me use hua; modern human medicine me iska use largely obsolete hai, lekin veterinary antiepileptic context me ab bhi mil sakta hai. [Q29]

EXAM TRAP — Q25 — Sulfadiazine terminology

- ▶ Sulfadiazine synthetic sulfonamide antibacterial drug hai; strict/classical definition me ise natural-product “antibiotic” se alag rakha ja sakta hai.
- ▶ Source exam explanation Penicillin + Sulfadiazine dono ko antibiotic-type antibacterial options treat karke “more than one” key karta hai.

5. Toxicology, Food Processing & Organic Food

- Carbon monoxide hemoglobin ke saath carboxyhemoglobin banata hai aur oxygen ke comparison me bahut high affinity se bind karta hai; isi wajah se tissue hypoxia aur CO poisoning hoti hai. [Q30]
- Milk homogenization ka goal fat globules ko microscopic size me todna hai taaki cream layer separate na ho aur texture uniform rahe. Clove oil ka main active component Eugenol hai, jisme antiseptic aur analgesic properties hoti hain. [Q31–Q32]
- Jelly making ke liye high-pectin + adequate-acid fruits best hote hain; Guava pectin-rich hone ki wajah se suitable hai. Organic food production generally synthetic fertilizers aur most conventional pesticides ke restricted/avoided use ke standards follow karta hai. [Q33–Q34]

EXAM TRAP — Q30, Q31 & Q34

- ▶ Q30 me “irreversible complex” wording technically loose hai: CO–Hb binding bahut strong hai, lekin chemically reversible hai; oxygen therapy se CO dissociate ho sakta hai.
- ▶ Q31 source option centrifuge bolta hai, lekin true homogenization high-pressure homogenizer se fat globules break karti hai; centrifugation cream separation ke liye use hoti hai.
- ▶ Q34 me organic = automatically safer/more nutritious prove nahi hota. Correct exam idea production method hai: synthetic fertilizers/pesticides par restrictions.

✓ Source-question mapping preserved: Q1–Q34